

Envy-Free Traffic Control for City-Wide Emergency Routing via Hierarchical Multi-Agent Actor–Critic

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Abstract

Efficient and equitable traffic signal control is critical for large-scale urban networks, particularly under emergency scenarios where routing priority must be dynamically enforced without degrading overall system performance. While hierarchical reinforcement learning (HRL) frameworks such as HiLight effectively balance global coordination and local adaptability, they rely on manually defined regional partitions and lack explicit mechanisms for emergency prioritization and fairness.

The proposed framework is trained using a hierarchical actor–critic paradigm, where a high-level policy generates region-level guidance and low-level agents execute intersection control decisions under combined objectives. Experimental evaluations demonstrate that our approach improves emergency vehicle throughput while preserving fairness and overall traffic efficiency, outperforming baseline HRL methods in complex urban scenarios.

Keywords: Traffic Signal Control, Multi-Agent Reinforcement Learning (MARL), Emergency Routing, Community Detection, Urban Traffic Optimization, Fairness-Aware Optimization

1 Introduction

Urban traffic congestion represents one of the most pressing infrastructure challenges of the modern era, with cascading consequences for emergency response efficiency, economic productivity, and public health. Among the various components of urban traffic management, the coordinated control of traffic signals at intersections and the rapid, unimpeded passage of emergency vehicles (EMVs) — including ambulances, fire trucks, and police vehicles — stand out as problems of acute societal importance. The urgency is quantified starkly by medical literature: the survivor rate from sudden cardiac arrest without treatment drops 7–10% for every minute elapsed, and survival probability approaches zero beyond eight minutes [2]. Yet despite decades of research, existing frameworks for traffic signal control and emergency vehicle management remain siloed, individually limited in scale, and largely blind to fairness considerations and real-world disruption scenarios.

Limitations of Existing Approaches

EMVLight [2] represents a landmark contribution to the problem of joint EMV routing and traffic signal pre-emption. Its core innovation is the formulation of each intersection as a reinforcement learning (RL) agent operating under a multi-agent advantage actor-critic (MA2C) framework, where agents are classified into three types — primary pre-emption, secondary pre-emption, and normal agents — depending on their proximity to the dispatched EMV. The spatial discount factor α in the adjusted reward function ensures that each agent pays proportionally less attention to the rewards of agents farther away, enabling decentralized execution without sacrificing global coordination awareness.

The decentralized dynamic Dijkstra routing heuristic further allows each intersection to update its estimated time of arrival (ETA) to the destination using only locally available information, reducing communication overhead compared to centralized re-routing. On synthetic and real-world benchmarks including a Manhattan 16×3 network and Hangzhou 4×4 , EMVLight achieves up to a 42.6% reduction in EMV travel time and a 23.5% reduction in average non-EMV travel time compared to existing methods.

However, EMVLight exhibits several important limitations that motivate further research. First, the framework is designed for *single EMV dispatch*: the reward structure and agent-type classification assume exactly one EMV in transit at any given time, and the pre-emption hierarchy collapses when multiple EMVs are simultaneously active. Second, the routing mechanism, while dynamic, remains fundamentally *reactive*: ETA updates at each intersection depend on observed intra-link travel times at the moment of update, and the framework has no capacity to anticipate congestion downstream of the EMV’s current position before it manifests.

Third, the network partitioning in EMVLight is implicit and unstructured — agents coordinate via neighborhood communication without any higher-level regional abstraction — meaning that global traffic patterns are invisible to individual agents beyond their immediate neighbors. Fourth, and critically, EMVLight provides *no fairness guarantees*: the pressure-based reward design optimizes for aggregate network

efficiency, but individual intersections that are structurally disadvantaged by their position in the network may receive systematically inferior service without any mechanism for remediation. Finally, EMVLight assumes a fixed network topology and does not address accident-induced capacity reduction, which is precisely the scenario most likely to trigger an EMV dispatch.

HiLight [1] addresses the scalability challenge that EMVLight does not directly confront. By introducing a hierarchical reinforcement learning (HRL) framework with two distinct policy levels — a global Meta-Policy and a decentralized Sub-Policy — HiLight achieves meaningful coordination across networks as large as the Manhattan2668 scenario with 2,668 intersections.

The Meta-Policy employs a Transformer encoder to perform information exchange across subregions and an LSTM to model temporal dependencies in regional traffic states, generating a global sub-goal vector that is broadcast to all Sub-Policy agents as a shared directional objective. The adversarial training mechanism, wherein the Meta-Policy generates increasingly challenging sub-goals while the Sub-Policy is trained to surpass them, provides a principled mechanism for joint optimization.

Nevertheless, HiLight carries its own limitations. First, the network partitioning into subregions is performed using a *fixed, manual grid-based approach*: for the Manhattan2668 network, a 4×15 grid is used.

4×15 grid is applied, producing 60 regions with approximately 45 intersections each. This partitioning is geometrically motivated but topologically uninformed — it does not respect the actual road network structure, meaning that community boundaries may cut through densely connected subgraphs and group intersections that share little traffic exchange. The resulting regional state representations may therefore be noisy or misleading for the Meta-Policy’s Transformer encoder. Second, HiLight assumes a *fixed number of neighbors* (exactly four) and a *fixed phase structure* (eight phases per intersection) in its Sub-Policy’s Graph Attention Concat (GAC) module. Real urban road networks, including city maps such as Coimbatore, contain irregular intersections with varying numbers of incoming links, lane configurations, and permissible movements — a three-way junction, a five-arm roundabout, and a standard four-way intersection each require different phase sets. HiLight’s fixed structural assumptions make direct transfer to such heterogeneous real-world networks non-trivial. Third, HiLight contains *no mechanism for emergency vehicle prioritization*: it is a general traffic signal control framework with no agent-type classification, no pre-emption logic, and no integration with EMV routing. Fourth, like EMVLight, HiLight provides *no fairness mechanism* at the intersection level and no treatment of supply-side disruptions such as accidents. Fifth, HiLight’s evaluation, while large in scale, remains confined to simulated environments with demand-side traffic variability; real-world deployability on an actual city road network with heterogeneous intersection geometries, multiple simultaneous EMVs, and stochastic incident injection has not been demonstrated.

The Present Work: Contributions and Overview

The framework proposed in this paper — addresses the above gaps through a set of coherent, technically motivated extensions to the hierarchical actor-critic paradigm.

Our contributions are as follows. **(1) Topology-Aware Community Detection via Louvain Algorithm.** We replace HiLight’s manual grid-based partitioning with an automated, topology-aware community detection procedure based on the *Louvain algorithm* [?]. The Louvain method is a greedy modularity optimization algorithm that partitions a graph into communities by iteratively reassigning nodes to the community that produces the greatest increase in modularity Q , defined as:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

where A_{ij} is the adjacency matrix of the road network graph, k_i and k_j are the degrees of nodes i and j , m is the total number of edges, and $\delta(c_i, c_j)$ is an indicator function that equals 1 if nodes i and j belong to the same community and 0 otherwise.

Applied to the traffic network graph—where intersections are nodes and road segments are weighted edges (with weights reflecting traffic volume or travel time)—the Louvain algorithm produces communities that correspond to densely connected sub-graphs with relatively sparse inter-community links. These communities represent natural traffic catchment zones: vehicles generated within a community predominantly circulate within it, crossing community boundaries primarily for longer trips.

We compare the Louvain algorithm against the *Leiden algorithm* [?], a refinement that addresses the connectivity guarantee weakness of Louvain by ensuring that all communities in the output partition are internally connected. While Leiden produces theoretically well-connected communities, our empirical evaluation on the road networks of Manhattan and Coimbatore city demonstrates that the Louvain algorithm consistently achieves higher modularity scores and more stable community assignments across random initializations. Leiden communities are smaller and more fragmented in the context of road networks, where the graph is naturally sparse and planar, reducing the effective inter-community coordination range of the Meta-Policy. We therefore adopt Louvain partitioning as the default configuration in all experiments, with a comparative analysis of modularity, normalized mutual information (NMI), and downstream traffic control performance presented in Section ???. The use of community-based partitioning has implications beyond computational convenience. Community boundaries in road networks tend to align with major arterials, river crossings, and other natural barriers — precisely the locations where inter-regional traffic management is most critical and where EMV corridor pre-clearance must be coordinated. The Meta-Policy operating over Louvain communities therefore has a structurally more meaningful representation of regional traffic state than a geometry-based grid, enabling more effective sub-goal generation. **(2) Dynamic Intersection Topology and Adaptive Phase Assignment.** Unlike HiLight, which fixes the number of neighbors at four and the signal phase set at eight for all intersections, our framework adopts a fully dynamic topology model. Each intersection agent i

Each intersection i is characterized by its actual number of incoming and outgoing links $\{l_i^{\text{in}}, l_i^{\text{out}}\}$, as determined by the road network topology. The permissible phase set $\mathcal{A}_i$ is generated automatically from the set of non-conflicting traffic movement combinations specific to the geometry of the intersection.

The Graph Attention module in the Sub-Policy is extended to handle variable-degree neighborhoods: attention scores are computed over $|\mathcal{N}(i)|$ neighbors (where $|\mathcal{N}(i)|$ varies across intersections), and zero-padding is applied only during feature aggregation rather than being embedded into the network architecture.

This design enables direct application to heterogeneous urban networks—including the road map of Coimbatore, Tamil Nadu, which contains a mixture of three-way junctions, four-way intersections, rotary intersections, and flyover merge points—without requiring manual preprocessing or network simplification.

(3) Hierarchical Multi-Agent EMV Routing with Multi-EMV Support.

We integrate EMVLight’s multi-class agent design into the hierarchical framework, extending it to support multiple simultaneously active EMVs. Each active EMV induces a dynamic agent-type classification over the intersections in its downstream corridor; when two EMV corridors overlap, a priority resolution mechanism arbitrates pre-emption sub-goals based on EMV type (ambulance, fire, police) and estimated remaining travel time. The Meta-Policy is extended with an EMV awareness module that identifies affected communities along each active EMV’s projected route and issues corridor pre-clearance sub-goals in anticipation of the EMV’s arrival — transforming EMV routing from a reactive, link-by-link operation to a proactive, community-level planning problem.

(4) Envy-Free Intersection-Level Fairness.

We introduce an envy-free regularization term into the Sub-Policy’s Actor-Critic loss, ensuring that no intersection agent systematically prefers the resource allocation received by a neighboring agent over its own. This mechanism is particularly important in the presence of EMV pre-emption, which creates temporary but significant disparities in green-time allocation between pre-emption agents and normal agents. The envy-free penalty encourages faster post-preemption recovery and produces a more equitable steady-state signal control policy.

(5) Accident-Driven Traffic Control on Coimbatore City Map.

We validate the complete framework on a real urban road network extracted from Coimbatore, Tamil Nadu, under stochastic accident injection scenarios with multiple simultaneous EMV dispatches. To our knowledge, this is the first evaluation of a hierarchical RL-based TSC framework with integrated EMV routing on an Indian city road network, and the first to treat the joint problem of accident-induced congestion and EMV passage in a unified learning framework. The remainder of this paper is organized as follows. Section 3 reviews relevant literature. Section ?? presents the problem formulation and community detection methodology. Section ?? describes the proposed framework architecture. Section ?? details experimental setup and results. Section ?? concludes with directions for future work.

2 LITERATURE SURVEY

2.1 Background

The N-Body problem is concerned with the interactions between celestial bodies [?], where every body in the given system of bodies is affected by every other body. The creation of galaxies, the effects of black holes and even the search for the dark matter are concerned with the N-Body problem. They all use Newton’s law of gravitation

to show the effects on the motions of the bodies due to the forces of gravity acting between them.

This project focuses on the celestial N-Body problem; with bodies interacting due to Newton's Law of Gravitation. The N-Body problem can be described as follows: where $N \geq 2$; For every $i = 1, 2, \dots, N$; let B_i with mass m_i be at $\vec{x}_i$, with velocity $\vec{v}_i$ and acceleration $\vec{a}_i$ at time $t \geq 0$ and where r_{ij} is the distance between B_i and B_j ; $i \neq j$ and $r_{ij} = r_{ji} \neq 0$. The force acting on B_i due to B_j is $\vec{F}_{ij}$; a function of the distance between them (r_{ij}). Assume that $\vec{F}_{ij} = -\vec{F}_{ji}$. Given initial positions and velocities of all $B_i \forall i = 1, 2, \dots, N$; the general N-Body problem is to determine the motion of the system if each B_i interacts with all other B_j 's in the system.

2.2 Literature Survey

We have referred to cite EMVLight to understand the implementation of actor critic reinforcement learning agents to each intersection. These agents make sure that there is no congestion in their respective as well as neighboring intersections. This is achieved by taking a reward function which utilizes the rewards of the neighbors, with a parameter alpha where the farther an intersection is the lower its reward affects the current intersection. wrote a small brief actual info can be extended

HiLight is used to understand the how GNNs can be used in the traffic management problem. It utilizes adversarial GNNs so there are two policies - MetaPolicy and SubPolicy. The subpolicy tries to be as good or better than the metapolicy. This make ssure that the model performs well. The intersections utilize the actor critic feame-work as mentioned above. Need to add more info - Embeds the info of positional - Utilizes LSTM to better remember relevant information

Referred to another paper Actor-Critic. This was opted for as the EMVLight was deterministic. Whereas this was continuous.

In 1994, the Virgo Consortium was founded for the purpose of cosmological simulations on supercomputers. The largest problem worked on through the Virgo Consortium has been the Millennium Run [?]. It was created to model the formation of the universe and to track how galaxies and black holes are created. The simulation traced around 10 billion particles where each particle represented around 20 million galaxies. It used a code called *GADGET*– GALaxies with Dark matter and Gas intER-act [?], which was originally written in serial to model collisions between galaxies but has since been developed to run in parallel and is now used to model a larger range of astrophysical problems.

The Virgo Consortium produces the *GADGET* code along with *MPI-HYDRA* and *FLASH*. Although all codes are used to solve astrophysical N-body problems, they are used to simulate different areas. *MPI-HYDRA* looks at galaxy formation, star formation and the x-ray transmission due to the heating and cooling of galaxies. *FLASH* simulates the thermonuclear flashes seen on the surface of compact stars such as neutron stars and white dwarfs.

3 Future Works

3.1 Emergency Vehicle Routing

Current approaches to emergency vehicle (EMV) routing — including EMVLight — treat route optimization as an incremental update problem over pre-computed paths derived from Dijkstra’s algorithm. While EMVLight’s decentralized dynamic Dijkstra heuristic represents a meaningful improvement over static A* routing, it still operates reactively: the estimated time of arrival (ETA) at each intersection is updated based on observed intra-link travel times, but the framework cannot anticipate network-wide disruptions that have not yet propagated to the EMV’s immediate neighborhood. In complex urban scenarios — particularly those involving multiple simultaneous emergencies, sudden congestion cascades, or infrastructure failures — this myopic update strategy can lead to suboptimal route selection.

Our proposed extension addresses this limitation by integrating EMV routing directly into the hierarchical actor-critic structure introduced in this work. Rather than relying on a purely reactive heuristic, the high-level Meta-Policy will be extended to function as a *proactive route planner*: given its global view of regional traffic states across the entire network, it can anticipate downstream congestion before the EMV encounters it and issue sub-goals that pre-clear corridors along the most promising future routes. This is a qualitative departure from EMVLight’s design, where the secondary pre-emption agent is responsible for clearing only the immediately upcoming link.

Concretely, at each Meta-Policy decision step, a candidate route set will be generated based on current regional pressure readings and the known destination. The Meta-Policy will then issue sub-goals to Sub-Policy agents along each candidate corridor, instructing them to reduce queue lengths on those links in anticipation of EMV arrival. This transforms EMV routing from a two-horizon problem (current and next link) into a multi-horizon planning problem more naturally suited to the hierarchical framework. The adversarial training mechanism from HiLight will be retained: the Meta-Policy will set increasingly ambitious corridor-clearance targets, while Sub-Policy agents learn to meet or exceed those targets under real traffic dynamics.

Additionally, we note that the community detection approach used to partition the traffic network into regions — using the Louvain algorithm, which demonstrated superior modularity and clustering metrics compared to Leiden in our preliminary experiments — provides a natural structure for routing corridor selection. Because Louvain communities correspond to densely connected subgraphs with sparse inter-community edges, inter-community transitions represent natural bottlenecks for EMV passage. The Meta-Policy can therefore reason about routing at the community boundary level, issuing coordinated pre-emption sub-goals for a sequence of community transitions rather than link-by-link decisions. This is computationally far more tractable than full network-level planning while still providing genuinely global routing awareness — a property neither EMVLight nor HiLight currently achieves.

3.2 Envy Free Traffic Control

A dimension of fairness that existing traffic signal control (TSC) frameworks — including both EMVLight and HiLight — systematically neglect is *envy-freeness* at the intersection level. In economic theory, an allocation is envy-free if no agent prefers the allocation received by another agent over their own. Translated to the TSC context: an intersection agent i is *envy-free* if, given the current state of the network, it does not prefer the signal phase assignment and resource allocation (green time, priority weighting) received by any neighboring intersection j over its own.

This is not merely an aesthetic concern. In practice, intersection agents that receive systematically lower green-time allocations — typically because they serve lower-pressure roads or are downstream of congested links — accumulate structural disadvantages that propagate through the network. Vehicles waiting at chronically under-served intersections experience disproportionately long delays, creating localized pockets of inequity that aggregate reward functions mask. Pressure-based reward designs, such as those in EMVLight and HiLight, explicitly optimize for network-level flow smoothness but provide no guarantee that the resulting policy is equitable across intersections with heterogeneous demand profiles. Our proposed fairness extension introduces an *envy-free regularization term* into the Sub-Policy’s Actor-Critic loss.

The envy-free penalty $\mathcal{L}_{\text{envy}}$ is defined as the average pairwise envy across neighboring agents within each community, and is incorporated into the Sub-Policy loss as:

$$\mathcal{L}_{\text{envy}} = \frac{1}{|\mathcal{N}|} \sum_{(i,j) \in \mathcal{N}} \mathcal{E}_{ij}^t$$

where $\mathcal{N}$ denotes the set of neighboring agent pairs within the same community. The final Sub-Policy loss is given by:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{AC}} + \lambda_e \mathcal{L}_{\text{envy}}$$

The envy-free penalty $\mathcal{L}_{\text{envy}}$ is defined as the mean pairwise envy across all neighboring agent pairs within each Louvain community, and is incorporated into the Sub-Policy loss with a weighting coefficient λ_e .

Thus, the Sub-Policy agents are simultaneously incentivized to maximize their cumulative expected returns while minimizing the extent to which they would prefer the allocations of neighboring agents, resulting in a policy that is both efficient and locally equitable.

Crucially, this formulation is compatible with the hierarchical structure of our framework. The Meta-Policy continues to operate at the regional level and issues sub-goals based on global traffic objectives; the envy-free regularization operates entirely within the Sub-Policy and does not require modifications to the Meta-Policy’s architecture. The Louvain community boundaries serve a second purpose here: envy comparisons are computed *within* communities rather than globally, reducing computational overhead while ensuring that fairness is enforced where it matters most — among intersections that share traffic flows and compete for the same vehicle demand. In the context of EMV scenarios, envy-free control has a specific additional motivation. When an EMV passes through the network, primary and secondary pre-emption

agents receive systematically different reward signals from normal agents (as established in EMVLight’s multi-class agent design). Without a fairness mechanism, agents along the EMV’s corridor accumulate disadvantage relative to normal agents during the pre-emption period, and residual congestion effects persist after the EMV exits. The envy-free regularization will encourage faster restoration of balanced service across intersections following EMV passage, addressing the post-preemption congestion effect that neither EMVLight nor current hierarchical approaches explicitly manage.

3.3 Accidents based Traffic Control

A significant gap in existing TSC literature — including both EMVLight and HiLight — is the treatment of *unplanned, spatially localized disruptions* such as road accidents. HiLight evaluates under three realistic traffic flow conditions (Peak Transition, Adverse Weather, and Holiday Rush), all of which represent demand-side variability. However, none of these scenarios involve supply-side disruptions: sudden, partial, or complete loss of road capacity at specific locations due to incidents. EMVLight, similarly, focuses on the positive case of EMV passage without modeling the adverse case where an accident both generates the EMV dispatch *and* simultaneously degrades the road network the EMV must traverse.

Our proposed extension models accident-driven traffic control as a joint disruption-response problem on a real city map. We utilize the Coimbatore City road network, augmented with stochastic accident injection.

At each episode, one or more road segments are randomly selected according to an accident probability distribution derived from real crash report data, and their effective capacity is reduced by a configurable severity parameter $\kappa \in (0, 1]$, where $\kappa = 1$ denotes complete blockage.

The hierarchical framework must adapt to this setting in two ways. First, the Meta-Policy must detect the accident’s signature in regional traffic state observations — a sudden, localized pressure spike distinct from the gradual pressure buildup characteristic of peak-hour demand — and issue sub-goals that reroute vehicle flows around the affected region. Because the Louvain community structure partitions the network into subgraphs with high internal connectivity, the community containing the accident site will experience elevated internal pressure while neighboring communities will initially show little change. This provides a natural early-warning signal that the Meta-Policy can learn to exploit.

Second, the Sub-Policy agents adjacent to the incident must implement responsive signal phasing strategies that simultaneously serve diverted traffic (which may arrive from unexpected directions), accommodate any EMV dispatched in response to the accident, and maintain equitable service at unaffected intersections via the envy-free mechanism described above. This is a substantially harder coordination problem than either clean-network EMV routing or demand-driven congestion management, because the disruption is spatially abrupt, temporally evolving, and interacts with the EMV passage in a coupled manner.

References

- [1] Y. Zhu, H. Wen, G. Min, and M. Luo, HiLight: A Hierarchical Reinforcement Learning Framework with Global Adversarial Guidance for Large-Scale Traffic Signal Control, arXiv preprint arXiv:2506.14391, 2025.
- [2] Su, Haoran, et al. "EMVLight: A multi-agent reinforcement learning framework for an emergency vehicle decentralized routing and traffic signal control system." *Transportation Research Part C: Emerging Technologies* 146 (2023): 103955.